A 43.5dB Gain Unipolar a-IGZO TFT Amplifier With Parallel Bootstrap Capacitor for Bio-Signals Sensing Applications

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***Abstract*—In this paper, a high gain amplifier with phase compensation loop is presented. A structure of parallel gate cross-coupled transistors to both ends of differential pair drain and source is designed to improves the load impedance, which obtains sufficient gain and further reduces power consumption. A novel capacitor bootstrap load circuit is proposed. The capacitor bootstrap topology is constructed by the drain source resistance of the transistor working in the cut-off region, where the gate source parasitic capacitor of the transistor is in parallel with the bootstrap capacitor rather than the existing series structure, thereby only a small bootstrap capacitor is required. By avoiding the use of large capacitors, chip area can be effectively reduced without compromising performance such as gain and bandwidth. The amplifier is fabricated using 10- µm n-type a-IGZO TFT technology. Measurement results show that the proposed amplifier achieves a voltage gain of 43.5 dB and a common mode rejection ratio of 61.2 dB while maintaining low power consumption. The amplifier also exhibits a -3 dB bandwidth covering 0.4∼2.1KHz, encompassing major bioelectric frequency bands. A real-time ECG signal was successfully captured using the fabricated TFT amplifier and gel electrodes. It has great potential in flexible sensing and acquisition applications such as electro cardiogram (ECG), electro encephalogram (EEG), pulse detection, and other wearable applications.**

***Index Terms*—a-IGZO TFT, bio-signal amplifier, bootstrap load, cross-coupled structure, Electrocardiogram (ECG), frequency compensation.**

I. Introduction

HUMAN's continuous pursuit of life quality drives the development and progress of wearable electronic products. Existing wearable devices are mainly based on CMOS amplifiers, which have advantages such as stable closed-loop gain, excellent bandwidth, and good noise performance. However, wearable devices based on silicon-based circuits are mainly made up partly or entirely of rigid components, which have mismatch with the human body in mechanical properties, and long-term wear is easy to cause discomfort and motion artifacts, that affect signal quality, especially in large area array sensing applications. Thin-film transistor integration technology is increasingly recognized as a key driving force in wearable electronic applications with special requirements for wearing comfort and mobility flexibility due to its low cost for disposable products, lightweight, ultra-thin, and especially its ability to be bound to flexible substrates and integrate with flexible electrodes [1]. Among the numerous materials, amorphous indium gallium zinc oxide (a-IGZO) thin film transistor (TFT) is considered as one of the best choices to realize the wearable flexible front-end system for human body, due to its series of advantages including relatively high mobility, large area uniformity, simple manufacturing process, good bias stability [2], small size [3], high yield and high integration [4].

As for a wearable sensing front-end amplifier for monitoring or collecting human physiological signals, the biological signals sent from the human tissue surface are very weak. For example, the amplitude of ECG signals is about 0.5-4mV, and the amplitude of EEG signals is about 5-300μV. Therefore, the gain performance of the amplifier is the primary focus of attention. Due to its high mobility, CMOS amplifiers can achieve a gain of over 40 dB [5], [6]. Meanwhile, high gain means low input reference noise, and the input reference noise voltage of silicon-based circuits can reach the level of $\mathrm{nV}/\sqrt{Hz}$. Unlike silicon-based CMOS processes, increasing gain in TFT integrated circuits is particularly challenging due to its low intrinsic gain and lack of mature complementary transistors. On the other hand, due to gain limitations, the input reference noise of TFT amplifiers cannot reach the level of silicon-based amplifiers. However, the application focus of TFT technology and CMOS silicon-based technology is different: CMOS has the characteristics of high gain, low noise, etc., which is suitable for scenarios with high signal accuracy such as implantable neural signal front-end or very weak signal detection front-end. TFT amplifiers can be integrated with flexible electrodes due to their process characteristics. In the future, it can be directly stacked and integrated on the back of flexible electrodes to achieve integrated flexible sensors, effectively suppressing motion artifacts and improving signal-to-noise ratio by minimizing length of the connection between electrodes and circuits [7]. Although the accuracy

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is not comparable to silicon-based circuits, the current gain and bandwidth levels of TFTs circuits can also meet many amplification needs of major wearable bioelectric signals. In addition, TFT can be manufactured at low temperatures, and the cost of TFT circuits is lower than that of CMOS circuits, which is conducive to large-scale biological signal sensing and the widespread construction of disposable sensing products attached to the skin.

The basic TFT amplifier structure is a unipolar transistor "NMOS-Diode" [8], [9], [10], [11] topology, which has the advantages of large input and output swing, insensitivity to mismatch and external condition variability, but its major drawback is low gain, usually only amplifying the signal 3∼5 times. The existing methods to improve the gain of unipolar transistor amplifier can be summarized as positive feedback [12], [13], [14], [15], [16], capacitor bootstrap [17], [18], [19], [20], [21], [22], [23], also known as AC coupling structure in some literatures, and cascade amplification [10], [11], [18], [24], [25], [26], [27]. In the positive feedback method, the drain of the differential input transistor is connected to the source of the unit gain diode load, and its drain is cross coupled to the gate of another pair of differential load transistors. Although its gain has increased to around 20 dB, the static operating point of the structure is difficult to set, and the introduction of additional current sources also increases the power consumption of the system. Existing capacitor bootstrapping method is to construct an RC high-pass loop for the load of the circuit. In the load, the drain-source capacitance of the transistor working in the cut-off region is in series with the bootstrap capacitance, and the bootstrap capacitor of sufficient size is needed to obtain sufficient cutoff frequency, so the layout realization area is large. The voltage gain realized by this method is about 20 dB too. Most of the existing multi-stage amplifiers use "NMOS-Diode" amplifiers with one-stage gain around 10 dB to cascade, the gain stage usually does not exceed three stages, the total gain is within 35 dB, yet the loop stability after the multi-stage amplifier cascade has not been improved. It is worth noting that if such circuits are used as closed-loop applications, further consideration of phase margin issues is needed.

In order to solve the above problems, this paper presents an amplification topology with unipolar transistors. To achieve high gain and further improve CMRR, the amplifier adopts two gain stages cascaded fully differential structure. In the first gain stage, a load impedance enhanced structure is designed to obtain sufficient gain and reduce power consumption by parallel gate cross coupled transistors between the drain and source ends of the differential pair. In the second gain stage, a novel capacitor bootstrap load circuit is proposed. The capacitor bootstrap topology is constructed by the drain source resistance of the transistor working in the cut-off region, while the gate source parasitic capacitor is in parallel with the bootstrap capacitor. Owing to this topology, only a small bootstrap capacitor is needed, thus avoiding the use of large capacitance, and effectively controlling the chip area without reducing the performance of circuit gain and bandwidth. In addition, the compensation method of the neutralization capacitor is proposed to change the original right half plane zero into left half plane zero and improve the phase margin of the cascade circuit. The proposed amplifier with parallel bootstrap capacitor and frequency compensation using unipolar a-IGZO TFT technology was implemented. The tested results show that it achieves high gain and phase margin under the condition of low power consumption, suitable for the future more advanced flexible and wearable electronics.

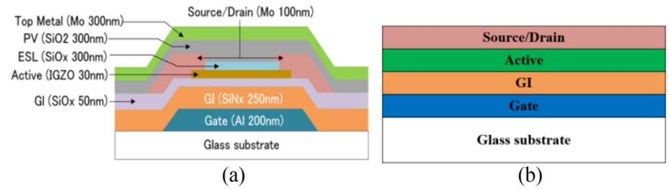

Fig. 1. (a) Structure of the a-IGZO TFT transistor, (b) structure of the a-IGZO TFT capacitor.

The remainder of this paper is organized as follows. Section II briefly introduces the process technology and device characteristics of the a-IGZO TFTs. The design and implementation of the proposed amplification topology are illustrated in Section III. In Section IV, the experimental results are presented and compared with the state-of-the-art. Finally, concluding remarks are provided in Section V.

II. Process Technology and Device Characteristic

A. Process Technology

In this work, a low-temperature a-IGZO process technology with etch-stop-layer (ESL) TFT is utilized for the development of the proposed amplifier on a glass substrate instead of a plastic substrate due to the simpler process and lower cost, as shown in Fig. 1(a). The device is fabricated by depositing different layers on top of a glass substrate. The stack consists of three metal layers (gate, active, top metal), an a-IGZO layer, dielectric layers (GI, ESL) and a passivation layer (interlayer). With a maximum process temperature of 350°C, this a-IGZO technology is also applicable to various plastic substrates for flexible electronics applications.

The detailed process of preparing ESL TFTs on a glass substrate is as follows. Firstly, on the glass substrate, a 200 nm thick, 20 µm-long aluminum (Al) layer was deposited by physical vapor deposition (PVD) sputtering as the gate electrode and patterned by wet etching. Secondly, the stacked grid insulator (GI) layer containing 250 nm thick SiO2 and 50 nm thick SiNx was deposited by plasma enhanced chemical vapor deposition (PECVD) and patterned by dry etching, and then the IGZO film prepared by radio frequency magnetron sputtering was deposited and patterned as the active layer with a thickness of 30 nm and a length of 15 µm. Next, 300 nm thick, 10 µm long SiOx layer was used as the etching stop layer (ESL), which prevented damage of the IGZO when processing the other layers. Then, Mo with a thickness of 100 nm was deposited by PVD sputtering and patterned into TFT with source and drain metal layers. Finally, 300 nm thick SiO2 layer was deposited by PECVD as the passivation layer of the protective device, and then 300 nm thick Mo was deposited and patterned as the top metal layer for circuit connection. The gate to source/drain overlap (OL) is 6.5 µm. In addition to the active devices, passive elements such as

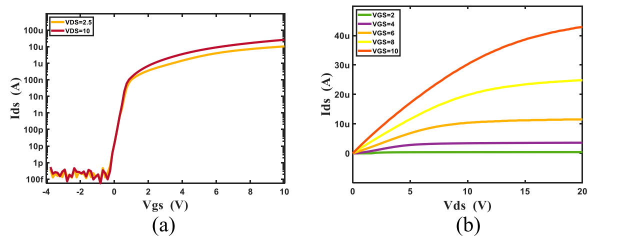

Fig. 2. (a) Measured transfer characteristics of IGZO TFTs with W/L = 20/10 μm. (b) Measured output characteristics of IGZO TFTs with W/L = 20/10 μm.

capacitors are also provided, the metal-insulator-metal (MIM) structure formed by the two metal layers exhibits a capacitance density of 16.5 nF/cm^2. The upper plate of the capacitor is composed of a source drain metal layer, and the lower plate is composed of a gate metal layer, While the gate insulation layer and active layer act as intermediate insulators, as shown in Fig. 1(b).

When source and drain metal is used as the source and drain electrode, it only needs to form contact with the active layer and only needs to open contact holes for ESL. Taking the TFT in the Fig. 1 above as an example, after depositing the ESL layer, it is necessary to open contact holes at two positions away from L_ESL when patterning the ESL layer. Afterwards, when source and drain metal layer is deposited, ESL through holes and contacts are naturally formed at the open positions.

When source and drain metal layer is used as a circuit interconnect, the active layer is not included below the source and drain. If the connection between source and drain and gate is to be formed, both GI and ESL need to open holes at the same designated position above the gate. When depositing source and drain metal layer, a through hole will be formed at that designated position.

B. Device Characteristic

Fig. 2 plots the typical transfer and output curves of the a-IGZO TFTs with a channel of 20 μm width and 10 μm length. The typical devices exhibited field-effect mobility of 12 cm^2/V·s, which was extracted by measuring the slope of I_{DS} versus V_{GS} at the deep linear region.

The measured transistor transfer characteristic curve shows that the leakage current during reverse bias is less than about 15pA during zero bias. When Vgs is reversed to below about -0.5 V, the Vds changes with the signal input, but the magnitude of leakage current does not exceed 0.5pA. This characteristic is conducive to the subsequent design of cutoff resistances based on Vgs reversed bias.

III. Circuit Design

The core circuit diagram of the amplifier proposed in this paper is shown in Fig. 3.

It includes a two-stage gain stage, an output buffer stage, and a frequency compensation loop. The first stage ($M1$-$M7$) uses the load impedance enhancement technique to achieve a voltage gain of about 20 dB. The second stage ($M8$-$M14$) employs the proposed bootstrap structure to achieve a voltage gain of about 23 dB, while reducing the area by two times compared to the

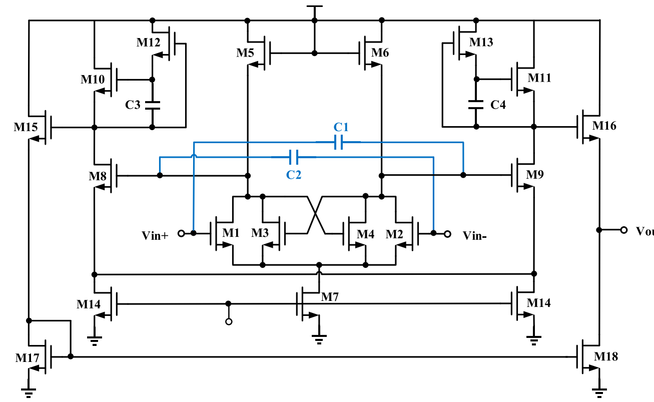

Fig. 3. Schematic of the proposed amplifier.

existing one. The output buffer stage ($M15$-$M18$) does not provide voltage gain and is responsible for converting the differential signal into a single-ended signal, improving the circuit current drive capability. Neutralizing capacitors ($C1$-$C2$) provide frequency compensation for the amplifier and ensure the stability of the high gain amplifier. The power consumption of the amplifier can be reasonably controlled by two tail current sources, $M7$ and $M14$. The total gain achieved by the amplifier is around 45 dB, about 200 times. Both gain stages are fully differential structures, so the common-mode rejection ratio of the amplifier can be expressed as follows:

$$CMRR = (1 + 2g_m R_{EE})\sqrt{\frac{R_D^2}{\Delta R_D^2} + \frac{g_m^2}{\Delta g_m^2}} \tag{1}$$

Where g_m is the transconductance of the differential input transistors, R_D is the load resistance, and R_{EE} is the output impedance of the tail current source. From (1) we can get the large output impedance of the tail current source contributing to high CMRR. In this process, the minimum channel length of the transistor can be up to 5 μm, but in order to ensure the yield and reliability of the amplifier and the large output impedance of the transistor, the channel length of the amplifier is designed to be 10 μm.

A. The Load Impedance Enhancement Technique

As shown in Fig. 4, the existing positive feedback structure requires five auxiliary transistors ($M6$-$M10$), which increases the circuit complexity and is susceptible to the mismatch variations of technology. The addition of the auxiliary current source $M10$ further increases the power consumption of the system. The power consumption of OPAMP designed based on this structure in reference [16] is 6.78mW.

The improved pseudo-CMOS structure of reference [28] is shown in Fig. 5. The circuit achieves low power consumption (160 μW) and 22.5 dB gain, but the phase margin measured by the tape out is about 20 degrees.

A cross-coupled based transconductance enhancement topology was proposed in literature [29], which achieves transconductance enhancement by introducing negative impedance at the source of the input transistors, as shown in Fig. 6. Based on this topology and positive feedback structure shown in [16], an

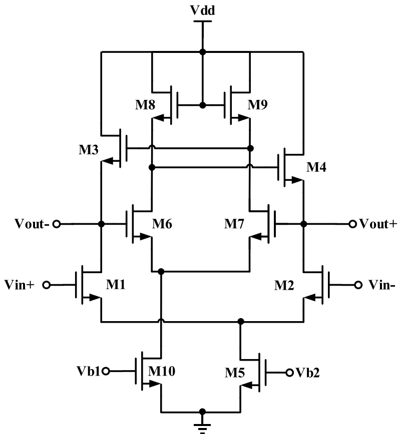

Fig. 4 Positive feedback structure.

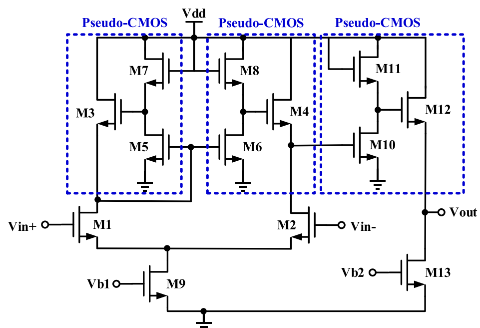

Fig. 5. Pseudo-CMOS architecture.

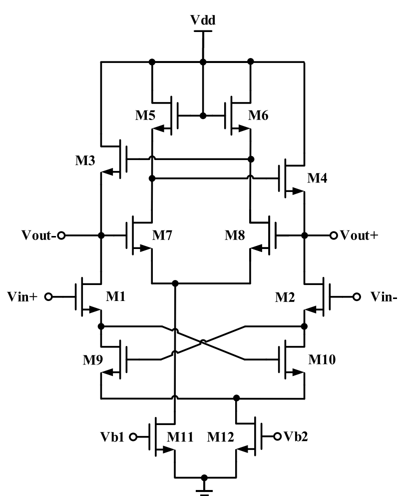

Fig. 6. Transconductance enhancement and positive feedback combined structure.

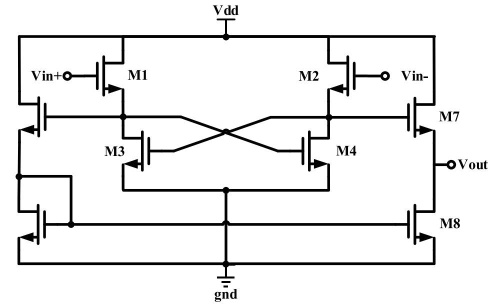

Fig. 7. Common drain amplification cross-coupled structure.

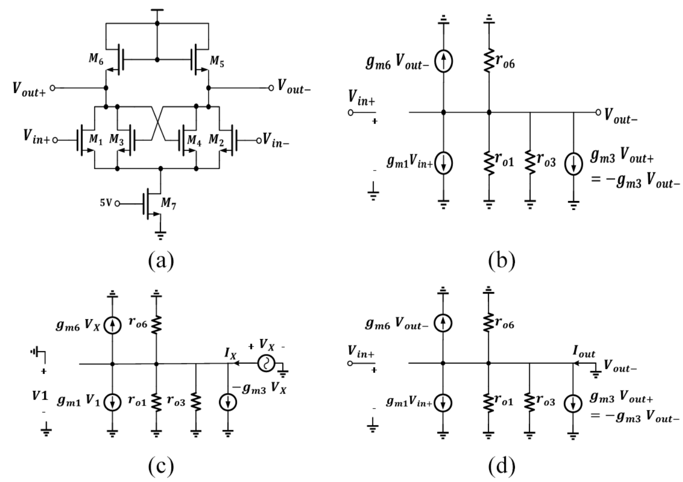

Fig. 8. (a) The load impedance enhanced structure, (b) small signal model of load impedance enhanced structure, (c) the equivalent circuit for calculating the output resistance, (d) the equivalent circuit for calculating the equivalent transconductance.

amplifier is designed. Due to the cross coupling structure introducing a negative resistance $R_S = -1/g_m$ [33], and when this negative resistance is located at the source, the equivalent transconductance of the input transistor is $G_M = g_{m1}/(1 + g_{m1}R_S) = g_{m1}/(1 - g_{m1}/g_{m9})$. When g_{m9} is set to be close to g_{m1}, a desirable equivalent transconductance can be obtained. But Two types of positive feedback loops are introduced to its gain stage, which affects the stability of the circuit if it will be used in closed-loop in the future. Moreover, a series cross-coupled module ($M9$ and $M10$) under the differential amplification transistor ($M1$ and $M2$) restricts the input common model range $V_{in,CM} \geq 2(V_{th} + \Delta)$. Where V_{th} is the turn-on voltage and Δ is the overdrive voltage.

Literature [30] proposes a common drain amplification cross-coupling structure as shown in Fig. 7. However, the gain stage of this structure does not contain a tail current source, and the pseudo-differential structure also affects the common-mode suppression performance of the amplifier.

In order to meet the requirements of high gain, high CMRR and low power consumption of the amplifier, the core circuit diagram based on the load impedance enhancement technology is shown in Fig. 8. The method is applied to the first gain stage of the amplifier.

Transistors $M1$ and $M2$ achieve differential amplification, transistor $M3$ and $M4$ constitute a cross-coupled module, transistors M5 and M6 are the load, and transistor M7 is the tail current source.

Unlike [29], the load impedance enhanced structure introduced a negative resistance at the drain of $M1$ and $M2$ through cross coupled structure, which is used to cancel the positive resistance of the NMOS-Diode loads. According to the small signal model, the load impedance of the first stage can be inferred as

$$R_{o1}=\frac{V_X}{I_X}=\frac{V_X}{\frac{V_X}{r_{o3}\|r_{o1}\|r_{o6}}+g_{m6}V_X-g_{m3}V_X}$$
$$=\left(\frac{1}{g_{m6}-g_{m3}}//r_{o1}//r_{o3}//r_{o6}\right) \tag{2}$$

We can see from (2), it is important to note that g_{m3} should be less than g_{m6} so that the load impedance would not become negative. Since the transconductance of the transistor operating in the saturation region can be expressed as $gm\approx\sqrt{2\mu C_{ox}\frac{W}{L}I_D}$, and from KCL we can get $I_6=I_1+I_3$. It is inferred that $(W/L)_3$ can be appropriately designed to be larger than $(W/L)_6$ to obtain a large R_{o1}. As a result, a higher gain could be acquired.

The gain of the first stage can be expressed as

$$A_{v1}=-g_{m1}R_{o1}=-g_{m1}\left(\frac{1}{g_{m6}-g_{m3}}//r_{o1}//r_{o3}//r_{o6}\right) \tag{3}$$

Compared with the conventional "NMOS-Diode" structure, (3) includes an additional $1/(g_{m6}-g_{m3})$ factor that is close to zero, resulting in improved gain performance. In comparison to the positive feedback structure shown in Fig. 4, the circuit employs three fewer transistors and does not introduce a new current source into the structure, making it more suitable for low power-consumption portable wearable devices. Note that although the gain of the first stage is increased to the intrinsic gain level due to the introduction of cross-coupled pairs, the gain is still relatively small due to the weak mobility of TFT devices, so a second stage should be introduced.

Additionally, compared to the positive feedback combined transconductance enhancement structure proposed in literature [29], the circuit only contains one positive feedback loop which promotes stability. Finally, compared with the common drain cross-coupling structure proposed in reference [30], this circuit features a tail current source that enhances CMRR.

In order for both $M3$ and $M4$ transistors to operate in the saturation region, it is necessary to ensure that $V_{out+}-V_{out-}\leq V_{th3,4}$, the turn-on voltage in this process is around 0.9 V, which still provides sufficient saturation margin for amplifying mV or even μV signals.

B. The Proposed Novel Bootstrap Load Structure

The proposed technique is applied to the second stage of the amplifier. The existing capacitor bootstrap structure and its small signal model are shown in the Fig. 9. It should be noted that in thin film transistor process, usually the gate source capacitance and gate drain capacitance are relatively large, and $M3$ works in the cutoff region equivalent to a cutoff resistance R_{off3} and a capacitor C_{gs3}.

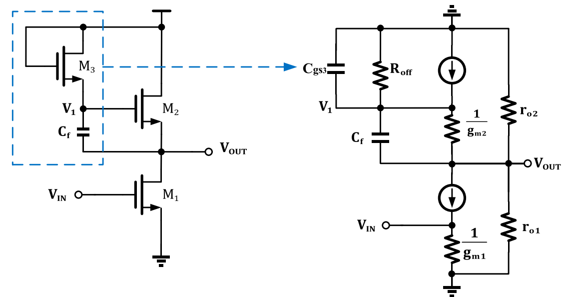

Fig. 9. Existing capacitor bootstrap structure and its small signal model.

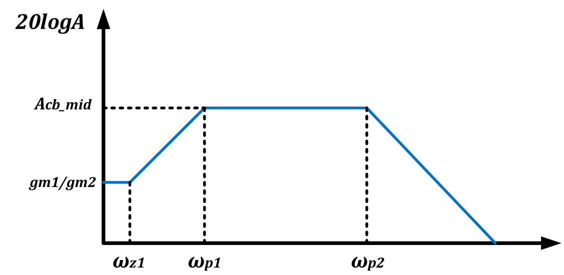

Fig. 10. Gain response of existing capacitor bootstrap structure.

The circuit transfer function can be expressed as

$$H_1(s)=\frac{g_{m1}\cdot(r_{o1}//r_{o2})[sR_{off3}(C_f+C_{gs3})+1]}{g_{m2}(r_{o1}//r_{o2})+1+s^2C_fC_{gs3}(r_{o1}//r_{o2})R_{off3}+s[(R_{off3}g_{m2}C_{gs3}+C_f)(r_{o1}//r_{o2})+R_{off3}(C_f+C_{gs3})]} \tag{4}$$

It can be obtained from (4) that the DC gain of the circuit is about g_{m1}/g_{m2}, the zero frequency of the circuit is

$$\omega_{z1}=\frac{1}{R_{off3}(C_f+C_{gs3})} \tag{5}$$

and the frequencies of poles are

$$\omega_{p1}=\frac{(r_{o1}//r_{o2})g_{m2}+1}{(R_{off3}g_{m2}C_{gs3}+C_f)(r_{o1}//r_{o2})+R_{off3}(C_f+C_{gs3})} \tag{6}$$

$$\omega_{p2}=\frac{(R_{off3}g_{m2}C_{gs3}+C_f)(r_{o1}//r_{o2})+R_{off3}(C_f+C_{gs3})}{C_fC_{gs3}(r_{o1}//r_{o2})R_{off3}} \tag{7}$$

The ratio of ω_{p1} to ω_{z1} is

$$\frac{\omega_{p1}}{\omega_{z1}}=1+\frac{g_{m2}(r_{o1}//r_{o2})R_{off3}}{\frac{C_{gs3}}{C_f}R_{off3}[1+g_{m2}(r_{o1}//r_{o2})]+R_{off3}+(r_{o1}//r_{o2})} \tag{8}$$

(8) indicates that ω_{p1} is always greater than ω_{z1}, so the gain response of the existing capacitor bootstrap structure is shown in Fig 10. The gain between frequencies ω_{p1} and ω_{p2} can be approximated as

$$A_{cb_{mid}}=g_{m1}\frac{(r_{o1}//r_{o2})}{g_{m2}(r_{o1}//r_{o2})+1}\frac{w_{p1}}{w_{z1}}$$
$$=g_{m1}\left[r_{o1}//r_{o2}//\frac{(1+C_f/C_{gs3})}{g_{m2}}\right] \tag{9}$$

At present, due to the capacitance value of C_{gs} being close to pF, the ratio of C_f to C_{gs} is not large enough in the process of thin film transistors, as shown in (9), resulting in a decrease in bootstrap gain. Therefore, a large C_f is required to quickly increase the gain to the ideal value of $A_{v\_ideal} = g_{m1}(r_{o1}//r_{o2})$ within the frequency band of interest.

In addition, due to the frequency of bioelectrical signals are generally relatively low, some even less than 1Hz, requiring a large C_f to get the desired bandwidth. For instance, supposing zero frequency of 0.05 Hz. According to measured results as shown in Fig. 2 combining with HSPICE simulation, $M3$ with W/L = 20/10 µm cutoff resistance R_{off3} is tens of GΩ above, which requires a C_f tens of pF above. Under the similar process size, it will probably occupy 0.6×2 mm^2 area including required transistors, which is not good for cost and high integration. In situations where high-resolution electrode monitoring is required, the minimum electrode spacing needed has strict requirements on the area, such as electromyography and electroencephalography.

In order to save chip area, the proposed core circuit diagram and small signal model of bootstrap structure designed in this paper are shown in Fig. 11. The circuit transfer function can be expressed as

$$H_2(s) = \frac{g_{m1}\cdot(\mathrm{r_{o1}//r_{o2}})[\mathrm{sR'_{off3}C'_f}+1]}{\begin{array}{c}g_{m2}(r_{o1}//r_{o2})+1+s^2C'_fC_{gd3}(r_{o1}//r_{o2})R'_{off3}\\ +s[(C_{gd3}+C'_f)(r_{o1}//r_{o2})+R'_{off3}C'_f]\end{array}} \tag{10}$$

Where $C'_f = C_f + C_{gs3}$ and the C_{gd3} is the gate drain capacitance of $M3$, The DC gain of the circuit is also g_{m1}/g_{m2}, the zero and poles frequencies are

$$\omega'_{z1} = \frac{1}{R'_{off3}C'_f} \tag{11}$$

$$\omega'_{p1} = \frac{(r_{o1}//r_{o2})g_{m2}+1}{(C_{gd3}+C'_f)(r_{o1}//r_{o2})+R'_{off3}C'_f} \tag{12}$$

$$\omega'_{p2} = \frac{(r_{o1}//r_{o2})(C'_f+C_{gd3})+R'_{off3}C'_f}{R'_{off3}C'_fC_{gd3}(r_{o1}//r_{o2})} \tag{13}$$

In the frequency band we are interested in, the gain can be approximated as

$$\begin{aligned}A_{cb2_{mid}} &= g_{m1}\frac{(r_{o1}//r_{o2})}{g_{m2}(r_{o1}//r_{o2})+1}\quad\frac{w_{p1}}{w_z}\\ &= g_{m1}\left[r_{o1}//r_{o2}//\frac{R'_{off3}C'_f}{C'_f+C_{gd3}}\right]\end{aligned} \tag{14}$$

As mentioned above, C_{gs3} in the conventional structure is in series with the bootstrap capacitor C_f, the ratio of C_f to C_{gs3} limits the voltage gain. However, C_{gs3} is in parallel with the bootstrap capacitor C_f in our proposed bootstrap structure, eliminating the constraint of the ratio of C_f to C_{gs3} on voltage gain. As shown in (14), the $R'_{off3}C'_f/(C'_f+C_{gd3})$ term is much larger than $r_{o1}//r_{o2}$ and can be ignored. Therefore, it does not require a large C_f to achieve the desired gain.

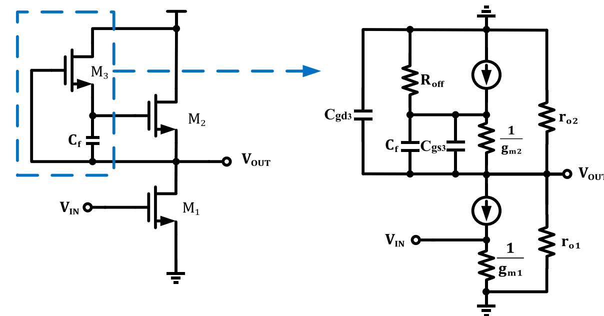

Fig. 11. Proposed bootstrap structure and small signal model.

On the other hand, assuming that the low-frequency cutoff frequency of the existing capacitor bootstrap structure and the proposed capacitor bootstrap structure are equal, that is, let (12) be equal to (6), and after simplification, we can obtain the following equation

$$R_{off3}g_{m2}(r_{o1}//r_{o2})C_{gs3} + R_{off3}C_f \approx R'_{off3}C'_f \tag{15}$$

Noting that $g_{m2}(r_{o1}//r_{o2})C_{gs3}$ is approximate to C_f, so (15) can simplify as

$$2R_{off3}C_f \approx R'_{off3}C'_f. \tag{16}$$

In order to obtain a larger resistance, we constructed a reverse bias Vgs below 0.5V in the circuit, as shown in Fig. 2. After simulation, the R'_{off3} in the proposed bootstrap structure is approximately four times that of the traditional structure. As the R'_{off3} is larger than $2R_{off3}$, from (16), we can infer that under the same low-frequency cutoff frequency, the proposed structure can be achieved with a smaller bootstrap capacitor than the existing capacitor bootstrap structure. Without reducing performance such as circuit gain and low-frequency cutoff frequency, the use of large capacitance is avoided, effectively reducing the chip area.

To prove the feasibility of the above theory, the existing bootstrap structure and the proposed bootstrap structure are verified by simulation. The low frequency cutoff frequency was set at 0.05 Hz to ensure the bootstrap amplification of the signal within the frequency band of the main bioelectrical signal (frequency range of ECG signal is 0.05-100 Hz, frequency range of EEG signal is 0.5-150 Hz). Using the existing structure, a transistor of W/L = 10u/10u and a capacitor of 40 pF are required to construct the bootstrap topology. While using our proposed architecture, only a transistor of W/L = 10u/10u and a capacitor of 10 pF are needed to obtain the same gain and low-frequency cutoff frequency, with an area reduction of about two times.

C. The Proposed Neutralization Capacitor Compensation Loop

The frequency response is obtained by simulating the amplifier in the range of 1-10 MHz, as shown in Fig. 13, where the yellow curve is phase frequency curve before compensation, and the blue curve is phase frequency curve after compensation.

Before compensation, it can be derived from phase frequency curve in Fig. 13 that the system has a low frequency zero ω_{z0}, a

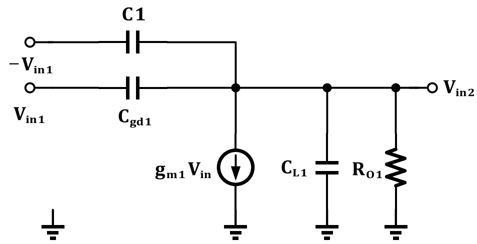

Fig. 12. The proposed compensation loop.

low frequency pole ω_{p0}, a main pole ω_{p1}, a second pole ω_{p2} and a high frequency zero ω_{z1}. According to the structural characteristics of the amplifier, it can be inferred that ω_{p0} and ω_{z0} are generated by the capacitor bootstrap stage, ω_{p1} is located at the output of the first stage, and ω_{p2} is located at the output of the second stage since the load capacitance of the first stage is larger than that of the second stage. The main pole ω_{p1} and the second pole ω_{p2} are given by the (17), (18).

$$\omega_{p1} \approx \frac{1}{\left(r_{o1}//r_{o3}//r_{o6}//\frac{1}{g_{m6}-g_{m3}}\right)C_{L1}} \tag{17}$$

$$\omega_{p2} \approx \frac{1}{(r_{o8}//r_{o10})C_{L2}} \tag{18}$$

Where C_{L1} and C_{L2} are the output capacitors of the first and second stages of the amplifier.

It should be noted that in thin film transistor process, usually gate source capacitance and gate drain capacitance are as relatively large. And the high frequency zero generated by the first stage is not far from the main pole, which can be derived

$$\omega_{z1} \approx \frac{g_{m1}}{C_{gd1}}(RHZ) \tag{19}$$

As can be seen from the blue curve Fig. 13, the high frequency zero ω_{z1} is close to the second pole ω_{p2}. Since it is the right half-plane zero, it cannot cancel out the second pole ω_{p2}, and the phase drops very fast, seriously affecting the stability of the circuit.

According to the proposed compensation loop, the input of the second stage provides the first stage with instant feedback. Specifically, the drain of transistor $M8$ is fed back to the gate of transistor $M1$ via capacitor C_1, and the drain of transistor $M9$ is fed back to the gate of transistor $M2$ via C_2. The equivalent circuit of half side of the first stage after adding compensation loop is shown in the Fig. 12. C_{L1} and R_{o1} represent the capacitance and resistance of the first stage output respectively.

It is known from KCL that

$$\begin{aligned}(V_{in1} - V_{out})sC_{gd1} - (V_{in1} + V_{in2})sC_1 \\ = g_{m1}V_{in1} + V_{in2}\left(sC_{L1} + \frac{1}{R_{O1}}\right)\end{aligned} \tag{20}$$

The transfer function of the system can be defined as

$$\frac{V_{in2}}{V_{in1}} = \frac{s(C_{gd1} - C_1) - g_{m1}}{s(C_{gd1} + C_1 + C_{L1}) + \frac{1}{R_{o1}}} \tag{21}$$

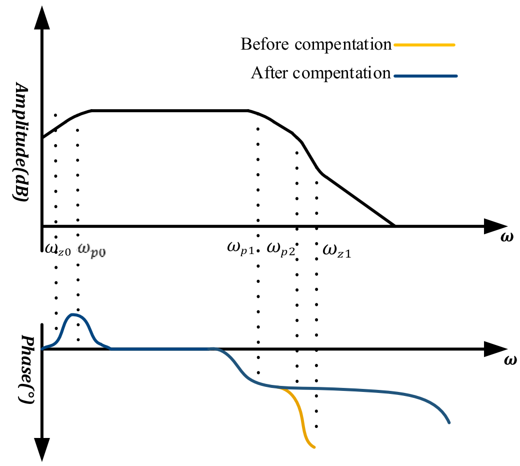

Fig. 13. Frequency response of the system before and after frequency compensation.

So, from (14), the high frequency zero ω'_{z1} after compensation can be described as

$$\omega'_{z1} = -\frac{g_{m1}}{C_1 - C_{gd1}}(LHZ) \tag{22}$$

Usually, the value of C_1 is greater than the C_{gd1}.Therefore, a right half plane zero can be converted to left half plane zero by means of the proposed neutralization capacitance compensation loop. It can be seen from Fig. 13 that the amplitude-frequency characteristics after compensation are basically unchanged, but the phase margin is greatly improved compared with that before compensation because the polarity of the original high-frequency zero is changed. The large phase margin ensures the stability of the amplifier when used in closed-loop, such as implementing a closed loop in wearable and other design gain control or controllable array detection applications to form a gain programmable front-end or intermediate stage gain programmable amplifiers.

IV. Measurement and Discussion

A. Circuit Characterization

The proposed amplifier designed using a-IGZO TFTs have been successfully fabricated on glass substrates. It is fully integrated on-chip and does not require external discrete components. Fig. 14 shows the micrograph of the amplifier with a 1.76×0.63 mm^2 area without considering the pad. The following test data are measured at 15 V power supply voltage. The tests use an oscilloscope (Pico Scope 5243D) with load of 10 MΩ//15 pF.

Fig. 15 shows the transient waveforms of the proposed amplifier. By means of waveform generator, a sinusoidal signal with the voltage peak–peak (Vpp) of 20 mV and frequency of 2 K Hz was applied to the input of the amplifier. Results show that the output Vpp of the amplifier is about 2 V.

Fig. 16 shows the measured Bode plot of the amplifier. In this measurement, two sinusoidal waves with Vpp of 20 mV and opposite phases were fed to the amplifier input. The amplifier

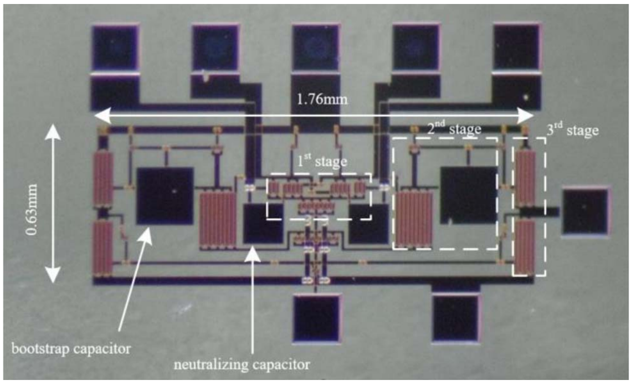

Fig. 14. Micrograph of the presented amplifier.

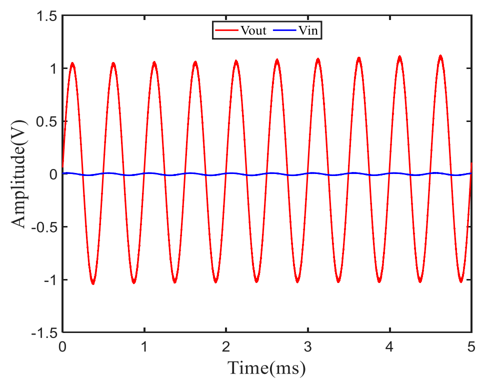

Fig. 15. Transient waveforms of the proposed amplifier

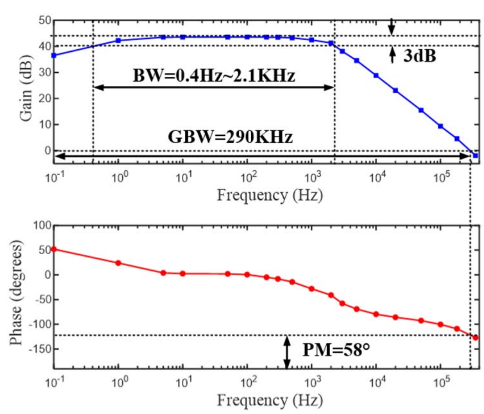

Fig. 16. Measured Bode plot of the amplifier.

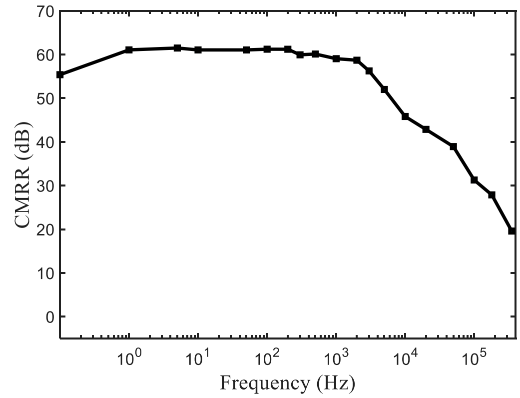

Fig. 17. Measured CMRR of the amplifier.

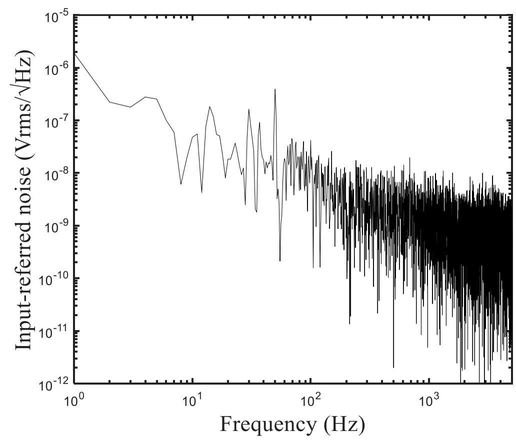

Fig. 18. Input-reference noise of the proposed amplifier.

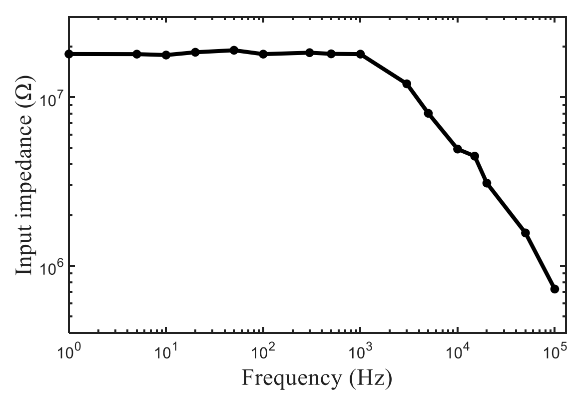

Fig. 19. Input impedance of the proposed amplifier.

shows a pass-band gain of 43.5 dB, a −3 dB bandwidth of 0.4 Hz–2.1 KHz, and a phase margin of 58 degrees.

The common mode rejection ratio (CMRR) is equal to the differential mode gain divided by the common mode gain. In order to obtain common mode gain, the same sinusoidal wave signal with Vpp of 100 mV was applied to the two input terminals of the amplifier when the corresponding output voltage was measured. Fig. 17 shows the measured CMRR of the amplifier. In the frequency band of interest, the CMRR is higher than 40 dB and can reach up to 61.2 dB.

In order to obtain the input-referred noise of the amplifier, the inputs of the amplifier were connected to small signal ground, and then we measured the power spectral density (PSD) of output voltage noise in the range of 1-5 kHz. Finally, we divided the obtained result by the amplifier gain at the corresponding frequency to obtain the power spectral density of the input-referred noise. The result is shown in Fig. 18.

Then integrate the obtained curve within the frequency band of interest to obtain the total input noise. The result shows that the input-referred noise value is 9.11 µVrms from 1 to 200 Hz and 12.67 µVrms from 1 to 2 KHz.

Due to the input bio-signals being divided by electrode impedance and input impedance, a high input impedance is

TABLE I
PERFORMANCE OF THE PROPOSED AMPLIFIER COMPARED TO THE STATE-OF-ART

Ref.	[31]	[32]	[13]	[19]	[18]	[27]	[23]	This Work
TFT Process	a-Si	a-IGZO	a-IGZO	Organic	Oxide	a-IGZO	Oxide	a-IGZO
Gain(dB)	20	30	24.9	35	22.8	23.5	20	43.5
BW(Hz)	0.06-200	150	4.9-5.2k	20	185-3.3k	3.7k	3-6k	0.4-2.1k
CMRR(dB)	30-50	-	-	44@50Hz	-	67.4@50Hz	42	61.2@100Hz
Input-referred noise (μVrms)	2.3(0-100Hz)	146(0-150Hz)	31.4(20-500Hz)	-	34.7(400-600Hz)	8(1-100Hz)	106(0-200Hz)	9.11(1-200Hz)
Power (mW)	11(55V)	0.188(20V)	1.3(±13V)	-	0.052(10V)	0.125(10V)	0.1(8V)	0.2(15V)
Input impedance (MΩ)	0.26	-	29.6	-	55.3	16.5	100	18
Area(mm^2)	90	-	11.2	200	37	123.4	41	1.1

required when used for obtaining bio-signals. Fig. 19 presents the measured input impedance of the amplifier.

The result shows that the input impedance of the proposed amplifier is above 8 MΩ from 1 Hz to 5 kHz, especially greater than 18MΩ within 1 kHz, which is high enough to minimize signal attenuation from the wet electrodes to the amplifier when used for obtaining bio-signals such as electrocardiogram (ECG) signals.

Table 1 summarizes the measurement results of the proposed amplifier and compared with previously reported results. The voltage gain achieved by the existing architecture is around 20-30 dB. this work achieves a gain of 43.5 dB with the above proposed technique while obtaining a phase margin of 58 degrees. Literatures [18], [23], [31] all adopt capacitor bootstrap structure, and the area of the circuit is large due to the existence of bootstrap capacitor. In this paper, the proposed novel capacitor bootstrap technique is adopted to design an amplifier with an area of 1.76×0.63 mm^2, and a higher CMRR is obtained by increasing the output impedance of the differential amplifier tail current source. With a power consumption of less than 200 μW, the amplifier is suitable for flexible wearable applications, and has advantages such as high gain, CMRR, and low power consumption, which can further extend the battery life of wearable devices.

B. In Vivo ECG Measurement

To demonstrate that the proposed TFT amplifier can be used for obtaining human biological signals, we have designed experiments directly targeting the human body to verify the amplification effect of actual bioelectrical signals. Fig. 20 shows the real scene of the measurement. The proposed TFT amplifier was connected to the interface board through a FPC cable, and three commercial Ag/AgCl wet electrodes were used. Two input electrodes were placed on the chest and directly connected to the input of the TFT amplifier through off-chip coupling capacitors, with a bias circuit behind the capacitors. The input bias circuit structure is as follows: Firstly, the input bias circuit uses MOS-diode transistors in series to divide the power supply voltage to the rated value, which is half of the power supply voltage due to no body effects in TFTs. Secondly, the bias voltage is applied to the gate of the input transistors through two TFTs working in the linear region. When the equivalent resistance provided by the bias circuit is connected in parallel with the input resistance of the input stage, an overall input impedance of approximately 18 MΩ can be obtained. Owing to the 18 MΩ impedance combined with 1uF coupling capacitors, the overall low-frequency cutoff frequency of the input is low enough,

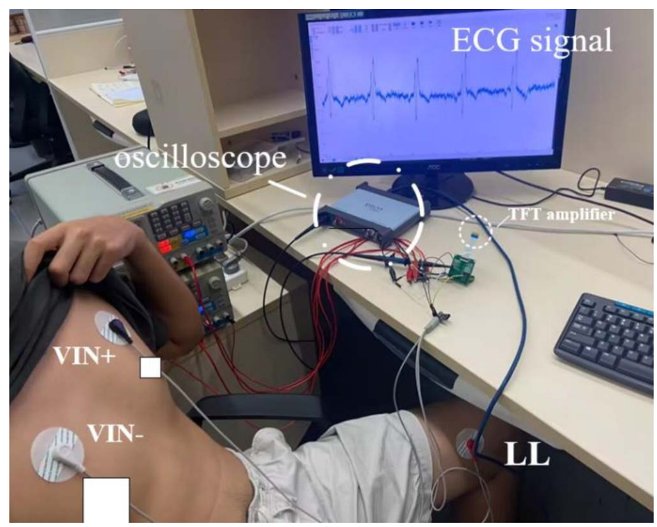

Fig. 20. In-vitro ECG measurement setup.

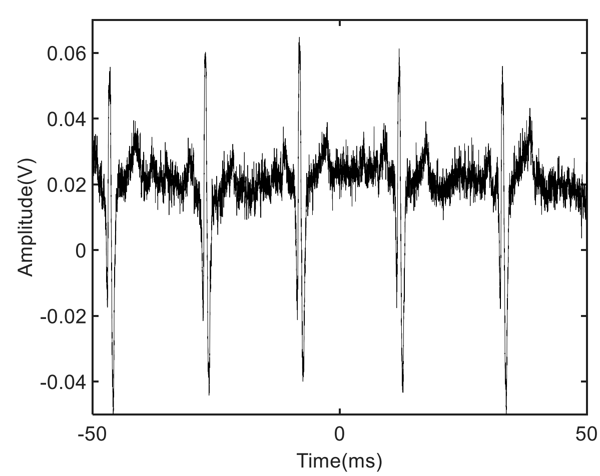

Fig. 21. The acquired ECG signal.

meeting the design needs. The LL electrode fed back the common mode level from the output of first stage to the human body. The output of the amplifier was directly connected to the oscilloscope and real-time waveform was displayed on the computer. Note that the data collected by the oscilloscope is reversed, so it is flipped in Fig. 21.

Fig. 21 shows the amplified ECG signal. The result shows that the Vpp of the output ECG signal is about 100 mV, and the corresponding Vpp of the input ECG signal is 1 mV. Thanks to the proposed technology, a relatively high gain and high common mode rejection ratio were obtained, resulting in a clear ECG signal.

V. Conclusion

This paper presents a novel amplifier topology adapted to unipolar process, which can be effectively utilized in TFT flexible biological signal sensing and acquisition systems. The proposed amplifier has two gain stages. In the first stage, parallel gate cross-coupled transistors are employed at both ends of the differential pair drain and source to enhance the load impedance, achieve sufficient gain, and reduce power consumption. The second stage incorporates a capacitor bootstrap topology that utilizes the drain-source resistance of a transistor operating in the cut-off region along with the gate-source parasitic capacitor connected in parallel with the bootstrap capacitor. This design approach minimizes reliance on large capacitance while effectively controlling chip area without compromising circuit performance in terms of gain and bandwidth. Furthermore, a neutralized capacitor compensation circuit is implemented to convert the original right half plane zero into left half plane zero, thereby enhancing circuit phase margin. Although all measurements are completed in open-loop to achieve maximum gain, a large phase margin can also ensure the stability of the amplifier when used in closed-loop applications. The proposed amplifier chip occupies only 1.76×0.63 mm^2 core area and exhibits -3 dB bandwidth covering 0.4 $\sim$ 2.1 KHz, 43.5 dB gain, and 61.2 dB common mode rejection ratio while maintaining low power consumption of 200 μW. Based on the technologies proposed above, a real-time ECG signal was successfully captured using the fabricated TFT amplifier and gel electrodes. Experimental results demonstrate the proposed amplifier has the advantages of high gain and great phase margin under low power conditions, and is in line with main bioelectrical signal frequency bands, making it suitable for future advancements in flexible and wearable electronics.

References

[1] T. He and C. Lee, “Evolving flexible sensors, wearable and implantable technologies towards BodyNET for advanced healthcare and reinforced life quality,” *IEEE Open J. Circuits Syst.*, vol. 2, pp. 702–720, 2021, doi: 10.1109/OJCAS.2021.3123272.

[2] M. Zulqarnain and E. Cantatore, “Analog and mixed signal circuit design techniques in flexible unipolar a-IGZO TFT technology: Challenges and recent trends,” *IEEE Open J. Circuits Syst.*, vol. 2, pp. 743–756, 2021, doi: 10.1109/OJCAS.2021.3123206.

[3] K. Myny, “The development of flexible integrated circuits based on thin-film transistors,” *Nat. Electron.*, vol. 1, pp. 30–39, Jan. 2018.

[4] J. Biggs et al., “A natively flexible 32-bit arm microprocessor,” *Nature*, vol. 595, pp. 532–536, Jul. 2021.

[5] K. Wen, S. Liu, L. Zhong, Y. Shen, and Z. Zhu, “A −64.3 dB THD, 26 nV/√ Hz bio-potential readout analog-front-end amplifier with a Gm-C integrator-implanted DC servo loop, and a bulk-driven ripple reduction loop,” *IEEE Trans. Circuits Syst. I, Reg. Papers*, vol. 71, no. 2, pp. 537–547, Feb. 2024

[6] P. Fath and H. Pretl, “A 370-nW quad-channel multi-mode bio-signal acquisition AFE with 2.9-μVrms input noise,” *in Proc. IEEE Nordic Circuits Syst. Conf. (NorCAS)*, Aalborg, Denmark, 2023, pp. 1–5.

[7] M. Sheeraz et al., “Flexible wearable biopatches for physiological monitoring using dry thin gold film electrodes,” *in Proc. IEEE Biomed. Circuits Syst. Conf. (BioCAS)*, Toronto, ON, Canada, 2023, pp. 1–5, doi: 10.1109/BioCAS58349.2023.10388742.

[8] N. Münzenrieder et al., “Flexible a-IGZO TFT amplifier fabricated on a free standing polyimide foil operating at 1.2 M Hz while bent to a radius of 5 mm,” *in Proc. Int. Electron Devices Meeting*, San Francisco, CA, USA, 2012, pp. 5.2.1–5.2.4, doi: 10.1109/IEDM.2012.6478982.

[9] J. Chang et al., “Fully printed electronics on flexible substrates: High gain amplifiers and DAC,” *Organic Electron.*, vol. 15, no. 3, pp. 701–710, 2014.

[10] K. Kim, K. Y. Choi, and H. Lee, “a-InGaZnO thin-film transistor-based operational amplifier for an adaptive DC–DC converter in display driving systems,” *IEEE Trans. Electron Devices, vol.* 62, no. 4, pp. 1189–1194, Apr. 2014.

[11] D. Kim, Y. Kim, K. Y. Choi, D. Lee, and H. Lee, “A solution-processed operational amplifier using direct light-patterned a-InGaZnO TFTs,” *IEEE Trans. Electron Devices*, vol. 65, no. 5, pp. 1796–1802, May 2018.

[12] Y.-C. Tarn, P.-C. Ku, H.-H. Hsieh, and L.-H. Lu, “An amorphous-silicon operational amplifier and its application to a 4-bit digital-to-analog converter,” *IEEE J. Solid-State Circuits*, vol. 45, no. 5, pp. 1028–1035, May 2010, doi: 10.1109/JSSC.2010.2043886.

[13] C. Garripoli, S. Abdinia, J.-L. J. P. van der Steen, G. H. Gelinck, and E. Cantatore, “A fully integrated 11.2-mm^2 a-IGZO EMG front-end circuit on flexible substrate achieving up to 41-dB SNR and 29-M Ω input impedance,” *IEEE Solid-State Circuits Lett.*, vol. 1, no. 6, pp. 142–145, Jun. 2018, doi: 10.1109/LSSC.2018.2878184.

[14] P. G. Bahubalindruni et al., *High-gain transimpedance amplifier for flexible radiation dosimetry using InGaZnO TFTs, IEEE J. Electron Devices Society*, vol. 6, pp. 760–765, 2018.

[15] R. Rodrigues, P. Bahubalindruni, and P. Barquinha, “High speed operational amplifier using a-InGaZnO TFTs with negative capacitance,” in *Proc. IEEE Int. Symp. Circuits Syst. (ISCAS), Piscataway, NJ, USA: IEEE Press*, 2020, pp. 1–5.

[16] R. Shabanpour et al., “A 70° phase margin OPAMP with positive feedback in flexible a-IGZO TFT technology,” in Proc. *IEEE 58th Int. Midwest Symp. Circuits Syst. (MWSCAS), Piscataway, NJ, USA: IEEE Press*, 2015, pp. 1–4.

[17] H. Fuketa et al., “1 μm-thickness ultra-flexible and high electrode-density surface electromyogram measurement sheet with 2V organic transistors for prosthetic hand control,” *IEEE Trans. Biomed. Circuits Syst.*, vol. 8, no. 6, pp. 824–833, Dec. 2014, doi: 10.1109/TBCAS.2014.2314135.

[18] M. Zulqarnain et al., “Design trade-offs in amorphous indium gallium zinc oxide thin film transistor based bio-signal sensing front-ends,” *Flexible Printed Electron.*, vol. 4, no. 1, 2019, Art. no. 014001.

[19] M. Sugiyama et al., “An ultraflexible organic differential amplifier for recording electrocardiograms,” *Nature Electron.*, vol. 2, no. 8, pp. 351–360, 2019.

[20] H. Fan et al., “A high gain low-noise amplifier based on ITO-stabilized ZnO thin-film transistors,” *IEEE Trans. Electron Devices*, vol. 67, no. 12, pp. 5537–5543, Dec. 2020.

[21] H. Marien, M. S. J. Steyaert, E. van Veenendaal, and P. Heremans, “Analog building blocks for organic smart sensor systems in organic thin-film transistor technology on flexible plastic foil,” *IEEE J. Solid-State Circuits*, vol. 47, no. 7, pp. 1712–1720, Jul. 2012, doi: 10.1109/JSSC.2012.2191038.

[22] P. G. Bahubalindruni et al., “Analog circuits with high-gain topologies using a-GIZO TFTs on glass,” *J. Display Technol.*, vol. 11, no. 6, pp. 547–553, 2014.

[23] Y. Xu et al., “Oxide TFT frontend amplifiers for flexible sensing systems,” *IEEE Trans. Electron Devices*, vol. 68, no. 12, pp. 6190–6196, Dec. 2021.

[24] C. Zysset, N. Münzenrieder, L. Petti, L. Büthe, G. A. Salvatore, and G. Tröster, “IGZO TFT-based all-enhancement operational amplifier bent

to a radius of 5 mm," *IEEE Electron Device Lett.*, vol. 34, no. 11, pp. 1394–1396, Nov. 2013, doi: 10.1109/LED.2013.2280024.

[25] M. Zulqarnain et al., "A 52 μW heart-rate measurement interface fabricated on a flexible foil with a-IGZO TFTs," *in Proc. IEEE 44th Eur. Solid State Circuits Conf. (ESSCIRC)*, Dresden, Germany, 2018, pp. 222–225, doi: 10.1109/ESSCIRC.2018.8494298.

[26] A. Rahaman et al., "A high performance operational amplifier using coplanar dual gate a-IGZO TFTs," *IEEE J. Electron Devices Society*, vol. 7, pp. 655–661, 2019.

[27] M. Zulqarnain, "A flexible ECG patch compatible with NFC RF communication," *npj Flexible Electron.*, vol. 4, no. 1, pp. 1–8, 2020.

[28] K. Ishida et al. "22.5 dB open-loop gain, 31 k Hz GBW pseudo-CMOS based operational amplifier with a-IGZO TFTs on a flexible film," in *Proc. IEEE Asian Solid-State Circuits Conf. (A-SSCC)*, Piscataway, NJ, USA: IEEE Press, 2014, pp. 313–316.

[29] Z. J. Chen et al., "A new high-gain operational amplifier using transconductance-enhancement topology integrated with metal oxide TFTs," *IEEE J. Electron Devices Society*, vol. 7, pp. 111–117, 2019.

[30] X. L. Mei et al., "A common drain operational amplifier using positive feedback integrated by metal-oxide TFTs," *IEEE J. Electron Devices Society*, vol. 9, pp. 342–347, 2021.

[31] T. Moy et al., "An EEG acquisition and biomarker-extraction system using low-noise-amplifier and compressive-sensing circuits based on flexible, thin-film electronics," *IEEE J. Solid-State Circuits*, vol. 52, no. 1, pp. 309–321, Jan. 2017, doi: 10.1109/JSSC.2016.2598295.

[32] C. Garripoli et al., "Analogue frontend amplifiers for bio-potential measurements manufactured with a-IGZO TFTs on flexible substrate," *IEEE J. Emerg. Sel. Topics Circuits Syst.*, vol. 7, no. 1, pp. 60–70, Mar. 2017, doi: 10.1109/JETCAS.2016.2616723.

[33] B. Razavi, "The cross-coupled pair—Part I [A circuit for all seasons]," *IEEE Solid-State Circuits Mag.*, vol. 6, no. 3, pp. 7–10, Summer 2014

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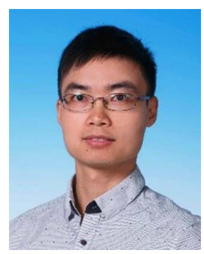

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